
Introduction

I transform organizations by understanding their business needs, creating clean data pipelines around those needs, and then doing data science to improve outcomes. I'm impactful as both a data engineer and a data scientist. I understand that doing meaningful data science requires having clean data. And I can communicate effectively at a business level.

Technical Stack

Languages & Frameworks: Python (Jupyter, Numpy, Pandas, Dash, SciPy), Clojure, Bash, R, Java (Micronaut, Spring)

Data & Infrastructure: Advanced SQL, Oracle, PostgreSQL (window functions, UDFs, PL/pgSQL, recursive CTEs), BigQuery, GCP (Google Logging, Pub/Sub, GCS, CX-Agents, Vertex AI, Composer), AWS, CI/CD, Jenkins, Apache Airflow, Kafka, Splunk, Spark, PySpark, Spark SQL, Docker, architecture (Kimball/star, snowflake, xNF)

Visualization Frameworks: Google Looker, Splunk, Dash, Plotly, R-Shiny, Seaborn, Matplotlib, Tableau

AI/ML/Stats: Vertex AI / GCP Conversational Agents, OpenAI, scikit-learn, TensorFlow, Monte Carlo, Bayesian inference, local LLMs, RAG, prediction transformer models, communicating stats effectively

Professional Experience

Rally Software	Data Engineer, Senior Data Engineer
CA Agile Central	Data Scientist, Senior Data Scientist, Principal Data Scientist
Broadcom (Rally – Agile Operations Division)	Staff Data Scientist
Broomfield, Denver, Boulder, CO	Oct 2015 – Oct 2025

- Led technical side of development for Agile flow metrics and Agile forecasting products.
- Developed a Smart Cumulative Flow Diagram for analyzing team and train delivery metrics.
- Led team and train-level portfolio flow metrics design and development.
- Used Vertex AI (Gemini) to produce flow metrics chart interpretations and coaching advice for teams and trains.
- Spearheaded capacity and delivery forecasting development for team and train-level planning.
- Developed, maintained, and used an idempotent distributed tracing lambda architecture data pipeline for a visible and reliable usage and performance metrics system to report 100% of user interactions in a SaaS environment involving 250K+ monthly users. Used the resulting data with Splunk, BigQuery, Looker, Spark, and other systems to analyze interactions. Held two years of long-lambda data for future needs. Data averaged 400GiB per day.
- Led performance optimization, via tracing and visualizations, for Rally's Oracle to PostgreSQL migration.
- Built and maintained a "product data hub" for those within Rally to track feature adoption, subscription usage, performance, and user input.
- Built a leading indicator for subscription renewal predictions using usage data and random forest models.
- Worked in Python, Java, Clojure, Bash, R, Scala, JavaScript (React), and elsewhere as needed.

SpotXChange
Broomfield, CO

Data Scientist
Apr 2014 – Sep 2015

Silicon Graphics, Inc. (SGI)
Longmont, CO

Software Engineer
Jun 2013 – Mar 2014

Education

Ph.D. in Mathematics – 2014/ABD

University of Colorado (Alexander Hulpke and Keith Kearnes)

- Computational group and graph theory, algebraic combinatorics, supercomputing

M.A. in Mathematics – 2010

University of Colorado (Keith Kearnes)

- Computational group theory, representation theory

M.S. in Mathematics – 2006

University of Vermont (David S. Dummit)

- Computational algebraic number theory

B.S. in Mathematics – 2004

University of Michigan–Flint

- Mathematics

About Me

Other Work

- I've taught many mathematics courses at the university level. I've taught several graduate level courses.
- Adjunct instructor: Saint Michael's College, Mathematics, 2006
- Adjunct instructor: Community College of Vermont, Photography, 2006
- Contract photographer: Automobile Magazine, location shoots, 2002-2005

Volunteering

- Over the winter holidays, I work with the Save Elephant Foundation and Elephant Nature Park in northern Thailand to rescue and rehabilitate abused elephants. My wife and I sponsor multiple elephants.
- I currently volunteer with A Precious Child in Broomfield, CO. I sort donations.
- Two-time week-long volunteer at the Mount Washington Observatory, including winter.

Random Stuff

- I've climbed 28 of Colorado's 14ers. Overall, I've had 39 14er summits.
- Since moving to Colorado, I've played in more than 750 softball games. I've played every position except pitcher. My favorite position is catcher, because nobody in D-league softball expects a good catcher.
- I've been to 51 countries.
- I rode bike trials in the Vans Warped Tour three times.
- I'm faceblind.